

PMBOK / PMI | PMB-30

Lessons Learned Register

Knowledge gained during the project, captured for use during it and after it

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Lessons Learned Register
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Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager
Approver	Not applicable — a living register
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-31	Project Closure Report	Consolidates these lessons
PMB-27	Issue Log	Issues often generate lessons
P3O-13	Lessons Repository	Organisational repository

How to use this template

Aspect	Guidance
Purpose	Records lessons as they arise so they can be applied within the project itself and passed to the organisation at closure. A lesson captured only at closure has already failed to benefit the project that learned it.
Framework	PMBOK / PMI — Integration Management — Manage Project Knowledge
Created when	From project start; maintained continuously
Reviewed / updated	At every phase boundary and at closure
Owner	Project manager
Approver	Not applicable — a living register
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	The recommendation matters more than the observation. "Communication was poor" is not a lesson; "hold a joint design session with the security team before, not after, architecture sign-off" is.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Each entry states the situation, what happened, why, and a specific recommendation
- Positive lessons are captured as well as negative ones
- Lessons are applied within the project where possible, and the application is recorded
- Lessons are categorised so they can be found by future projects
- At closure, lessons are transferred to the organisational repository with a named recipient

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1. Lessons register

Ref	Date	Phase	Category	Situation	What happened	Root cause	Recommendation	Applied in this project?	Owner
LL-001	12 Sep 2026	Tranche 1 pilot build	Planning and estimating	Tranche 1 pilot build	Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated VLAN	The security burn-in step was not represented as a distinct activity in the schedule	Model firmware verification as its own work package with parallel VLAN capacity sized for the shipment rate	Yes — reflected in the Tranche 2 schedule	S. Bhatt, Project Manager
LL-002	24 Sep 2026	Chargeback rate card	Governance and decision-making	Chargeback rate card	The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together	No single forum with both parties present and authority to decide	Convene a single decision workshop with Finance and business unit heads together, with the sponsor present to break deadlock	Yes — workshop held 24 Sep 2026	C. Whittfield, Finance Business Partner
LL-003	02 Oct 2026	Fabric commissioning	Technical and design	Fabric commissioning	Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass	Congestion control tuning is specialist work that the in-house team had not done before	Budget two weeks of vendor performance engineering per hall as a planned activity, not as a remedial response	Yes — included in the Tranche 2 estimate	J. Ferreira, Network Lead
LL-004	08 Sep 2026	Pilot penetration test	Security and compliance	Pilot penetration test	A critical finding on cluster API exposure was discovered late because	Assurance was treated as a gate rather than a	Run a security design review at architecture sign-off and a first penetration	Yes — assurance plan revised	L. Haddad, CI

Ref	Date	Phase	Category	Situation	What happened	Root cause	Recommendation	Applied in this project?	Owner
					the test was scheduled after build completion	continuous activity	test at 30% build, not only at completion		SO
LL-005	16 Sep 2026	Departmental cluster decommissioning	Stakeholder and communication	Departmental cluster decommissioning	Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced	The stakeholder register recorded them as supportive on the basis of a single early conversation	Assess attitude against observed behaviour, and engage those losing autonomy before the decision is communicated	Partially — engagement plan revised	F. Mansour, Business Change Lead
LL-006	05 Sep 2026	Integration partner award	Procurement and suppliers	Integration partner award	Awarding fabric and platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage	Evaluation optimised for commercial efficiency without weighting supplier concentration	Include supplier concentration explicitly in source selection criteria for packages above AED 10m	No — for the next procurement	E. Santos, Procurement Lead
LL-007	22 Jan 2027	Storage acceptance	Quality and testing	Storage acceptance	Benchmarking against a representative training workload rather than a synthetic test exposed a metadata bottleneck that synthetic tests missed	Synthetic benchmarks did not reproduce the small-file access pattern of the real corpus	Make workload-representative benchmarking a mandatory acceptance criterion for storage	Yes — reflected in acceptance criteria	I. Petrova, Data Platform Lead
LL-008	11 Sep 2026	SRE recruitment	Team and resourcing	SRE recruitment	Widening the role to remote-first within GST±3	The role was scoped to on-site	Test the necessity of location constraints	Yes	P. Raghav

Ref	Date	Phase	Category	Situation	What happened	Root cause	Recommendation	Applied in this project?	Owner
					produced four qualified candidates within three weeks after three months of local search	Dubai presence without testing whether that was necessary	before opening scarce technical roles		an, Platform Engineering Lead
LL-009	11 Aug 2026	Tenant isolation design	Requirements	Tenant isolation design	Involving the CISO at architecture option stage rather than at review removed two weeks of rework on the network policy model	Earlier programmes had treated security as a reviewer rather than a design participant	Include the security architect as a named participant in architecture option workshops	Yes	K. Adeyemi, Security Architect
LL-010	09 Sep 2026	Facilities and hardware fast-tracking	Planning and estimating	Facilities and hardware fast-tracking	Fast-tracking the structural order against a design that had not been frozen created a live risk of rework worth around AED 900,000	Schedule pressure at the Tranche 2 gate drove a decision to overlap design and procurement	Where fast-tracking is used, record the design freeze point explicitly and make change to it a Board decision	Yes — design freeze gate added	S. Bhatt, Project Manager

2. Lessons applied within the project

Lesson ref	Applied when	Change made	Effect observed
LL-001	Yes — reflected in the Tranche 2 schedule	Firmware verification added as a distinct work package in the WBS	Escalated to the Programme Board and resolved on 18 Sep 2026
LL-002	Yes —	Single decision workshop pattern added	Validated — no change required to the

Lesson ref	Applied when	Change made	Effect observed
	workshop held 24 Sep 2026	to the governance approach	baseline
LL-003	Yes — included in the Tranche 2 estimate	Two weeks of vendor performance engineering budgeted per hall	Not achieved; root cause analysed and an improvement item raised
LL-004	Yes — assurance plan revised	Penetration test moved to 30% build as well as completion	Delivered and accepted at the PI 4 system demo
LL-005	Partially — engagement plan revised	Attitude assessed from observed behaviour, not a single conversation	Partially met — carried into PI 5 with reduced scope
LL-006	No — for the next procurement	Supplier concentration to be weighted in source selection above AED 10m	Closed with no residual exposure

3. Lessons for the organisation

Complete at closure. Name the recipient — a lesson sent to a shared drive is not transferred.

Lesson ref	Recommendation	Relevant to	Transferred to	Date	Accepted?
Excluded from Tranche 2 and deferred to the AUH-2 second site	25 Jun 2027	Escalated to the Portfolio Board where it exceeds programme tolerance	Group Security	31 Mar 2027	No
Agreed with the integration partner at design freeze, 15 Oct 2026	28 May 2027	Confirmed with the FM provider under the existing facilities contract	Portfolio Board	26 Feb 2027	Yes
Assured independ ently by Cordant Assurance	23 Apr 2027	Covers the 2,048- GPU estate and the parallel filesystem tier	P3O Lessons Repository	31 Jan 2027	Partially
Carried forward from the Tranche 1	31 Mar 2027	Scoped to the three onboarding tenants: AI Research, Fraud	Finance Business Partner	15 Dec 2026	Yes

Lesson ref	Recommendation	Relevant to	Transferred to	Date	Accepted?
<i>lessons register</i>		<i>Analytics and Retail</i>			
<i>Enforced automatically in the delivery pipeline, with evidence retained</i>	26 Feb 2027	<i>Handled through the Agile Release Train rather than the predictive workstreams</i>	<i>Head of P3O</i>	30 Nov 2026	No
<i>Reviewed at each PI boundary by the Release Train Engineer</i>	31 Jan 2027	<i>Funded from the Tranche 2 allocation of AED 118,000,000</i>	<i>Group Security</i>	15 Oct 2026	Yes

4. Category analysis

Category	Count	Positive	Negative	Theme
Planning and estimating	3	<i>Enforced automatically in the delivery pipeline, with evidence retained</i>	<i>Applies across DXB-1 halls A and B for the Tranche 2 scale build</i>	<i>Applies from hall A handover; hall B follows in Tranche 2 closure</i>
Requirements	8	<i>Delegated to the Programme Manager within tolerance; above tolerance to the Programme Board</i>	<i>Delegated to the Programme Manager within tolerance; above tolerance to the Programme Board</i>	<i>Applies to all workloads on the DXB-1 estate without exception</i>
Stakeholder and communication	8	<i>Programme Board pack</i>	<i>Weekly workstream status return</i>	<i>Tranche 2 gate evidence pack</i>
Technical and design	3	<i>Reviewed monthly at the</i>	<i>Reflected in the solution</i>	<i>Confirmed with the FM provider under the existing facilities contract</i>

Category	Count	Positive	Negative	Theme
		Programme Board	intent as a fixed constraint	
Quality and testing	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks met in full
Procurement and suppliers	8	Nexora Integration Services	Gulf Mechanical and Electrical	Sable Talent Partners
Governance and decision-making	3	Approved with conditions	Deferred to the next Board	Approved with conditions
Security and compliance	3	Reported through the monthly benefit tracking report	Escalated to the Portfolio Board where it exceeds programme tolerance	Dependent on the Enterprise Identity Programme (DEP-003)
Team and resourcing	8	Platform services team	Infrastructure team	Observability team

5. Lessons review record

Date	Phase / event	Participants	Lessons added	Actions taken
12 Sep 2026	Tranche 1 pilot build	Programme Manager and workstream leads	Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated	Model firmware verification as its own work package with parallel VLAN capacity sized for the shipment rate

Date	Phase / event	Participants	Lessons added	Actions taken
			VLAN	
24 Sep 2026	Chargeback rate card	Programme Board	The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together	Convene a single decision workshop with Finance and business unit heads together, with the sponsor present to break deadlock
02 Oct 2026	Fabric commissioning	Agile Release Train — all teams	Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass	Budget two weeks of vendor performance engineering per hall as a planned activity, not as a remedial response
08 Sep 2026	Pilot penetration test	Service Operations and FM provider	A critical finding on cluster API exposure was discovered late because the test was scheduled after build completion	Run a security design review at architecture sign-off and a first penetration test at 30% build, not only at completion
16 Sep 2026	Departmental cluster decommissioning	Business Owners and tenant representatives	Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced	Assess attitude against observed behaviour, and engage those losing autonomy before the decision is communicated
05 Sep 2026	Integration partner award	CISO and Security Architect	Awarding fabric and	Include supplier concentration explicitly in source selection

Date	Phase / event	Participants	Lessons added	Actions taken
			<i>platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage</i>	<i>criteria for packages above AED 10m</i>